

LOSO (glove_ml_explorer) Experiment SA2

Static · full sensors (flex+ypr+accel, both hands), EXCLUDING Double_Nothing

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Experiment configuration

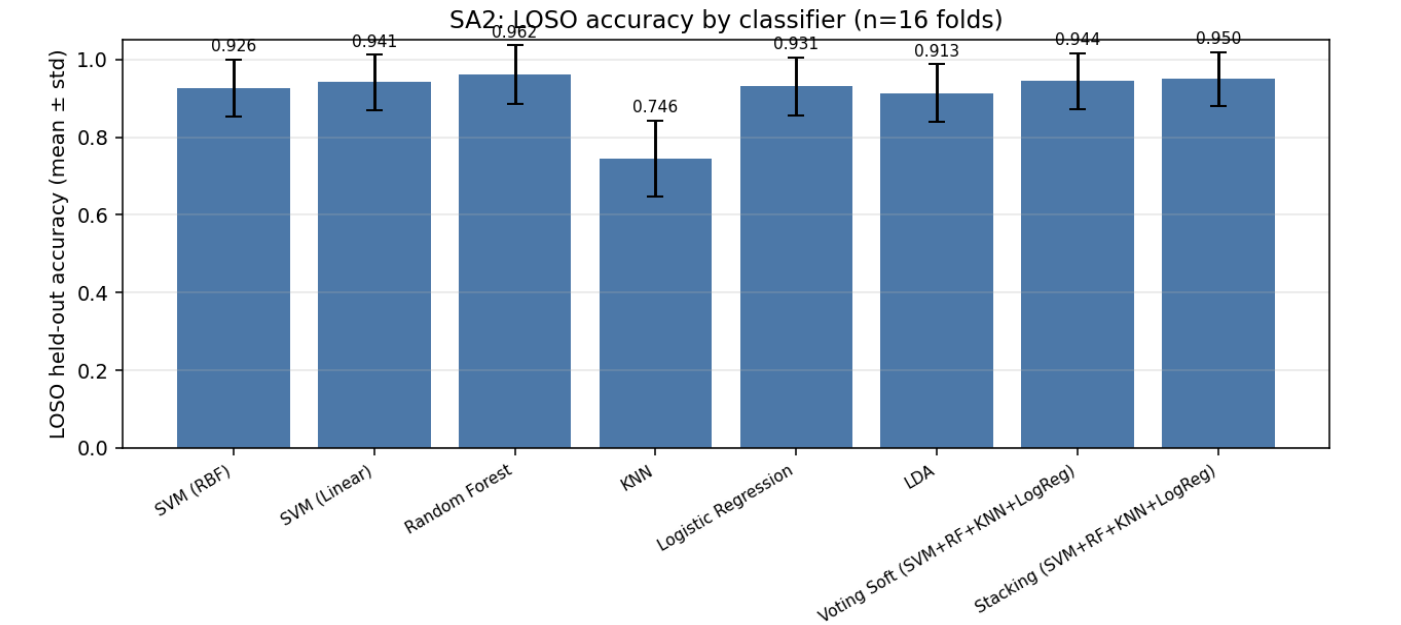
Class set	6 classes excluding Double_Nothing
Classes	Double_Closed_Fist, Double_Okay, Double_Open_Palm, Double_Phone, Double_Pistol, Double_Spiderman
Sensor modalities	flex, YPR, accel
Hands	left, right
Segments	thumb, index, middle, ring, pinky, wrist
Sensor channels	152
Sequence length	90 steps (resampled)
Filter	off
Feature mode	stats+fft
Normalisation	minmax
Augmentation	off
Inner CV folds	5
LOSO subjects	16: 10_Josh, 11_Mansh, 12_Marcus, 13_Nat, 14_StephenV2, 16_Bella, 17_Raquel, 1_Alan, 2_Alex, 3_Anghad, 4_Daniel, 5_Georgia, 6_Harry, 7_Henry, 8_Jestin, 9_Joselyn

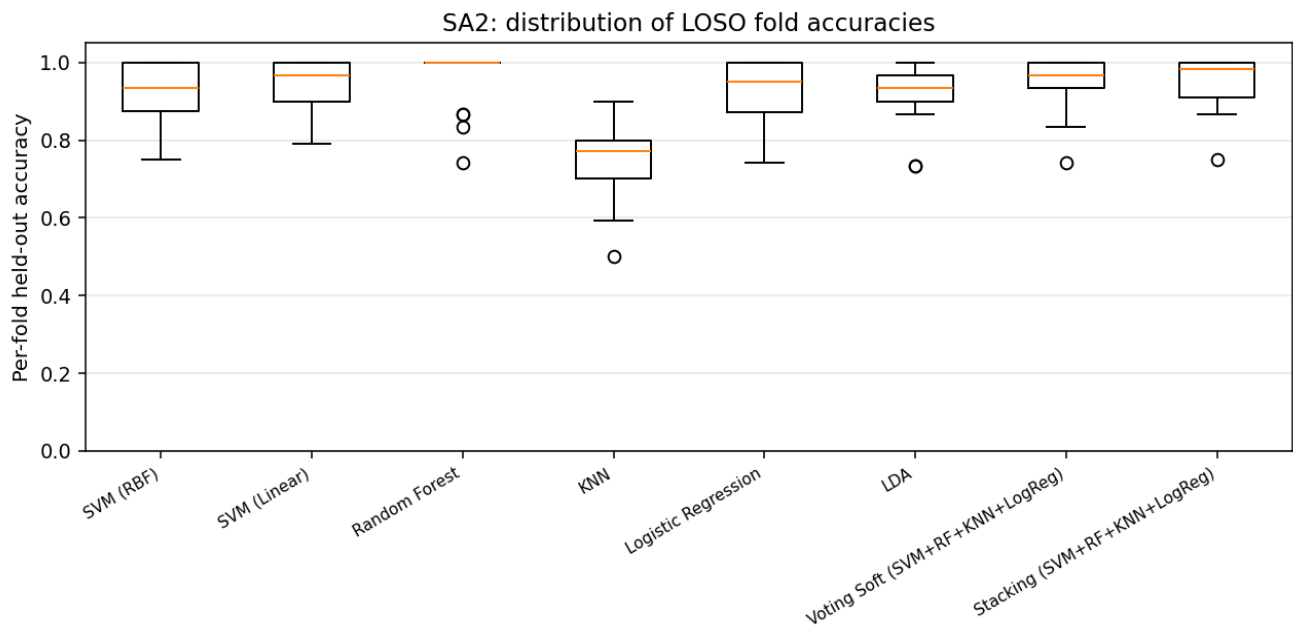
LOSO summary across all subjects

Classifier	Params*	LOSO mean	LOSO std	Min fold	Max fold	Inner CV mean
SVM (RBF)	—	0.9258	0.0739	0.7500	1.0000	0.9688
SVM (Linear)	123,135	0.9411	0.0706	0.7917	1.0000	0.9736
Random Forest	15,668	0.9616	0.0758	0.7417	1.0000	0.9748
KNN	—	0.7456	0.0974	0.5000	0.9000	0.7967
Logistic Regression	49,254	0.9308	0.0752	0.7417	1.0000	0.9694
LDA	49,254	0.9135	0.0747	0.7333	1.0000	0.9392
Voting Soft (SVM+RF+KNN+LogReg)	—	0.9438	0.0731	0.7417	1.0000	0.9777
Stacking (SVM+RF+KNN+LogReg)	—	0.9497	0.0689	0.7500	1.0000	0.9797

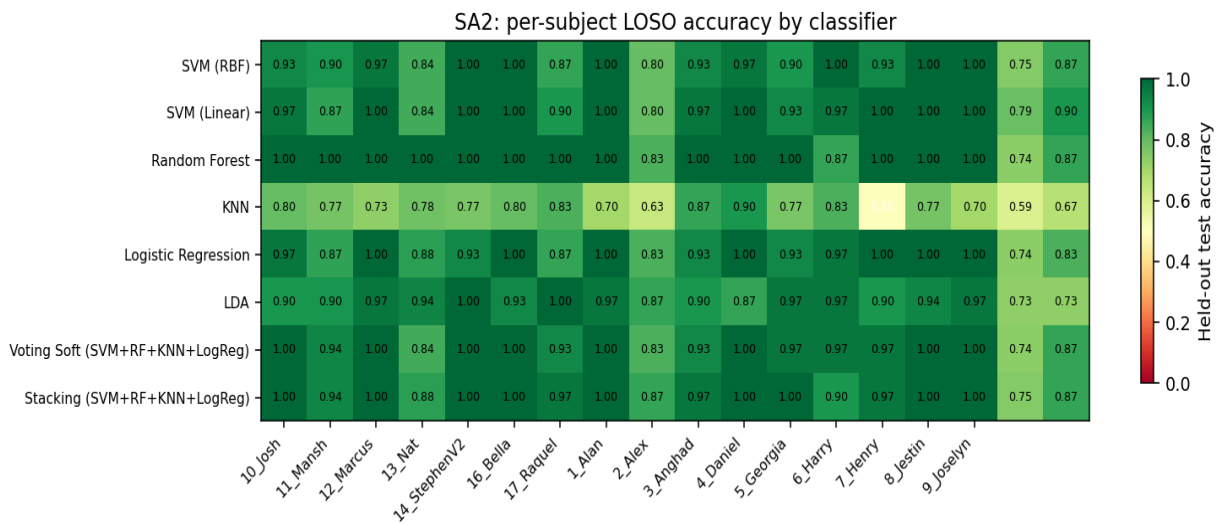
*Params: linear models use coef_+intercept size; tree ensembles use total node count across trees. SVM-RBF / KNN / Voting / Stacking are non-parametric or composite, so they show as —.

Best classifier by LOSO mean: **Random Forest**





Per-subject LOSO accuracy heatmap



Per-fold detail (held-out subject = each row)

Held-out subj	Train trials	SVM (RBF)	SVM (Linear)	Random Forest	KNN	Logistic Regression	LDA	Voting Soft (SVM+RF+KNN+LogReg)	Stacking (SVM+RF+KNN+LogReg)
10_Josh	604	0.933	0.967	1.000	0.800	0.967	0.900	1.000	1.000
11_Mansh	603	0.903	0.871	1.000	0.774	0.871	0.903	0.935	0.935
12_Marcus	604	0.967	1.000	1.000	0.733	1.000	0.967	1.000	1.000
13_Nat	602	0.844	0.844	1.000	0.781	0.875	0.938	0.844	0.875
14_StephenV2	604	1.000	1.000	1.000	0.767	0.933	1.000	1.000	1.000
16_Bella	604	1.000	1.000	1.000	0.800	1.000	0.933	1.000	1.000
17_Raquel	604	0.867	0.900	1.000	0.833	0.867	1.000	0.933	0.967
1_Alan	604	1.000	1.000	1.000	0.700	1.000	0.967	1.000	1.000
2_Alex	604	0.800	0.800	0.833	0.633	0.833	0.867	0.833	0.867
3_Anghad	604	0.933	0.967	1.000	0.867	0.933	0.900	0.933	0.967
4_Daniel	604	0.967	1.000	1.000	0.900	1.000	0.867	1.000	1.000
5_Georgia	604	0.900	0.933	1.000	0.767	0.933	0.967	0.967	1.000
6_Harry	604	1.000	0.967	0.867	0.833	0.967	0.967	0.967	0.900
7_Henry	604	0.933	1.000	1.000	0.500	1.000	0.900	0.967	0.967
8_Jestin	603	1.000	1.000	1.000	0.774	1.000	0.935	1.000	1.000
9_Joselyn	604	1.000	1.000	1.000	0.700	1.000	0.967	1.000	1.000
Stephen	514	0.750	0.792	0.742	0.592	0.742	0.733	0.742	0.750
Val	604	0.867	0.900	0.867	0.667	0.833	0.733	0.867	0.867

Classifier: SVM (RBF)

Params (best-effort): —

LOSO mean accuracy: 0.9258 ± 0.0739 (min 0.7500, max 1.0000)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	3/5	5/5	5/5	5/5	5/5	5/5	0.933
11_Mansh	3/6	5/5	5/5	5/5	5/5	5/5	0.903
12_Marcus	5/5	5/5	5/5	4/5	5/5	5/5	0.967
13_Nat	6/6	5/5	5/5	0/5	6/6	5/5	0.844
14_StephenV2	5/5	5/5	5/5	5/5	5/5	5/5	1.000
16_Bella	5/5	5/5	5/5	5/5	5/5	5/5	1.000
17_Raquel	5/5	2/5	5/5	4/5	5/5	5/5	0.867
1_Alan	5/5	5/5	5/5	5/5	5/5	5/5	1.000
2_Alex	4/5	5/5	5/5	0/5	5/5	5/5	0.800
3_Anghad	4/5	5/5	4/5	5/5	5/5	5/5	0.933
4_Daniel	5/5	5/5	5/5	4/5	5/5	5/5	0.967
5_Georgia	5/5	3/5	4/5	5/5	5/5	5/5	0.900
6_Harry	5/5	5/5	5/5	5/5	5/5	5/5	1.000
7_Henry	5/5	5/5	4/5	5/5	4/5	5/5	0.933
8_Jestin	6/6	5/5	5/5	5/5	5/5	5/5	1.000
9_Joselyn	5/5	5/5	5/5	5/5	5/5	5/5	1.000
Stephen	5/20	18/20	13/20	18/20	18/20	18/20	0.750
Val	2/5	5/5	5/5	4/5	5/5	5/5	0.867
MEAN	0.853	0.939	0.947	0.839	0.983	0.994	0.926

Classifier: SVM (Linear)

Params (best-effort): 123,135

LOSO mean accuracy: 0.9411 ± 0.0706 (min 0.7917, max 1.0000)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	4/5	5/5	5/5	5/5	5/5	5/5	0.967
11_Mansh	3/6	4/5	5/5	5/5	5/5	5/5	0.871
12_Marcus	5/5	5/5	5/5	5/5	5/5	5/5	1.000
13_Nat	6/6	5/5	5/5	0/5	6/6	5/5	0.844
14_StephenV2	5/5	5/5	5/5	5/5	5/5	5/5	1.000
16_Bella	5/5	5/5	5/5	5/5	5/5	5/5	1.000
17_Raquel	5/5	2/5	5/5	5/5	5/5	5/5	0.900
1_Alan	5/5	5/5	5/5	5/5	5/5	5/5	1.000
2_Alex	4/5	5/5	5/5	0/5	5/5	5/5	0.800
3_Anghad	4/5	5/5	5/5	5/5	5/5	5/5	0.967
4_Daniel	5/5	5/5	5/5	5/5	5/5	5/5	1.000
5_Georgia	5/5	4/5	4/5	5/5	5/5	5/5	0.933
6_Harry	5/5	5/5	5/5	4/5	5/5	5/5	0.967
7_Henry	5/5	5/5	5/5	5/5	5/5	5/5	1.000
8_Jestin	6/6	5/5	5/5	5/5	5/5	5/5	1.000
9_Joselyn	5/5	5/5	5/5	5/5	5/5	5/5	1.000
Stephen	7/20	18/20	16/20	18/20	18/20	18/20	0.792
Val	3/5	5/5	5/5	4/5	5/5	5/5	0.900
MEAN	0.881	0.939	0.978	0.861	0.994	0.994	0.941

Classifier: Random Forest

Params (best-effort): 15,668

LOSO mean accuracy: 0.9616 ± 0.0758 (min 0.7417, max 1.0000)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	5/5	5/5	5/5	5/5	5/5	5/5	1.000
11_Mansh	6/6	5/5	5/5	5/5	5/5	5/5	1.000
12_Marcus	5/5	5/5	5/5	5/5	5/5	5/5	1.000
13_Nat	6/6	5/5	5/5	5/5	6/6	5/5	1.000
14_StephenV2	5/5	5/5	5/5	5/5	5/5	5/5	1.000
16_Bella	5/5	5/5	5/5	5/5	5/5	5/5	1.000
17_Raquel	5/5	5/5	5/5	5/5	5/5	5/5	1.000
1_Alan	5/5	5/5	5/5	5/5	5/5	5/5	1.000
2_Alex	5/5	5/5	5/5	0/5	5/5	5/5	0.833
3_Anghad	5/5	5/5	5/5	5/5	5/5	5/5	1.000
4_Daniel	5/5	5/5	5/5	5/5	5/5	5/5	1.000
5_Georgia	5/5	5/5	5/5	5/5	5/5	5/5	1.000
6_Harry	5/5	5/5	5/5	1/5	5/5	5/5	0.867
7_Henry	5/5	5/5	5/5	5/5	5/5	5/5	1.000
8_Jestin	6/6	5/5	5/5	5/5	5/5	5/5	1.000
9_Joselyn	5/5	5/5	5/5	5/5	5/5	5/5	1.000
Stephen	4/20	18/20	20/20	11/20	18/20	18/20	0.742
Val	2/5	5/5	5/5	4/5	5/5	5/5	0.867
MEAN	0.922	0.994	1.000	0.864	0.994	0.994	0.962

Classifier: KNN

Params (best-effort): —

LOSO mean accuracy: 0.7456 ± 0.0974 (min 0.5000, max 0.9000)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	3/5	3/5	4/5	4/5	5/5	5/5	0.800
11_Mansh	3/6	3/5	5/5	3/5	5/5	5/5	0.774
12_Marcus	4/5	2/5	4/5	2/5	5/5	5/5	0.733
13_Nat	6/6	5/5	3/5	0/5	6/6	5/5	0.781
14_StephenV2	3/5	5/5	2/5	3/5	5/5	5/5	0.767
16_Bella	3/5	5/5	5/5	1/5	5/5	5/5	0.800
17_Raquel	5/5	3/5	5/5	2/5	5/5	5/5	0.833
1_Alan	5/5	4/5	2/5	0/5	5/5	5/5	0.700
2_Alex	5/5	1/5	2/5	1/5	5/5	5/5	0.633
3_Anghad	4/5	5/5	3/5	4/5	5/5	5/5	0.867
4_Daniel	5/5	5/5	5/5	2/5	5/5	5/5	0.900
5_Georgia	4/5	2/5	5/5	2/5	5/5	5/5	0.767
6_Harry	5/5	5/5	3/5	2/5	5/5	5/5	0.833
7_Henry	4/5	3/5	0/5	0/5	3/5	5/5	0.500
8_Jestin	6/6	5/5	3/5	0/5	5/5	5/5	0.774
9_Joselyn	3/5	4/5	3/5	1/5	5/5	5/5	0.700
Stephen	4/20	18/20	8/20	6/20	17/20	18/20	0.592
Val	1/5	2/5	5/5	3/5	5/5	4/5	0.667
MEAN	0.750	0.739	0.678	0.350	0.969	0.983	0.746

Classifier: Logistic Regression

Params (best-effort): 49,254

LOSO mean accuracy: 0.9308 ± 0.0752 (min 0.7417, max 1.0000)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	4/5	5/5	5/5	5/5	5/5	5/5	0.967
11_Mansh	3/6	4/5	5/5	5/5	5/5	5/5	0.871
12_Marcus	5/5	5/5	5/5	5/5	5/5	5/5	1.000
13_Nat	6/6	5/5	5/5	1/5	6/6	5/5	0.875
14_StephenV2	4/5	5/5	4/5	5/5	5/5	5/5	0.933
16_Bella	5/5	5/5	5/5	5/5	5/5	5/5	1.000
17_Raquel	5/5	1/5	5/5	5/5	5/5	5/5	0.867
1_Alan	5/5	5/5	5/5	5/5	5/5	5/5	1.000
2_Alex	4/5	5/5	5/5	1/5	5/5	5/5	0.833
3_Anghad	4/5	5/5	4/5	5/5	5/5	5/5	0.933
4_Daniel	5/5	5/5	5/5	5/5	5/5	5/5	1.000
5_Georgia	5/5	4/5	4/5	5/5	5/5	5/5	0.933
6_Harry	5/5	5/5	5/5	4/5	5/5	5/5	0.967
7_Henry	5/5	5/5	5/5	5/5	5/5	5/5	1.000
8_Jestin	6/6	5/5	5/5	5/5	5/5	5/5	1.000
9_Joselyn	5/5	5/5	5/5	5/5	5/5	5/5	1.000
Stephen	4/20	16/20	17/20	17/20	17/20	18/20	0.742
Val	1/5	5/5	5/5	4/5	5/5	5/5	0.833
MEAN	0.839	0.922	0.958	0.881	0.992	0.994	0.931

Classifier: LDA

Params (best-effort): 49,254

LOSO mean accuracy: 0.9135 ± 0.0747 (min 0.7333, max 1.0000)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	3/5	5/5	4/5	5/5	5/5	5/5	0.900
11_Mansh	4/6	4/5	5/5	5/5	5/5	5/5	0.903
12_Marcus	5/5	5/5	5/5	4/5	5/5	5/5	0.967
13_Nat	6/6	5/5	5/5	3/5	6/6	5/5	0.938
14_StephenV2	5/5	5/5	5/5	5/5	5/5	5/5	1.000
16_Bella	4/5	5/5	5/5	4/5	5/5	5/5	0.933
17_Raquel	5/5	5/5	5/5	5/5	5/5	5/5	1.000
1_Alan	5/5	5/5	5/5	4/5	5/5	5/5	0.967
2_Alex	4/5	5/5	5/5	3/5	4/5	5/5	0.867
3_Anghad	3/5	5/5	5/5	4/5	5/5	5/5	0.900
4_Daniel	4/5	5/5	4/5	3/5	5/5	5/5	0.867
5_Georgia	5/5	4/5	5/5	5/5	5/5	5/5	0.967
6_Harry	5/5	5/5	5/5	4/5	5/5	5/5	0.967
7_Henry	5/5	5/5	3/5	5/5	4/5	5/5	0.900
8_Jestin	5/6	4/5	5/5	5/5	5/5	5/5	0.935
9_Joselyn	4/5	5/5	5/5	5/5	5/5	5/5	0.967
Stephen	1/20	14/20	19/20	17/20	19/20	18/20	0.733
Val	3/5	3/5	4/5	2/5	5/5	5/5	0.733
MEAN	0.808	0.928	0.942	0.836	0.975	0.994	0.913

Classifier: Voting Soft (SVM+RF+KNN+LogReg)

Params (best-effort): —

LOSO mean accuracy: 0.9438 ± 0.0731 (min 0.7417, max 1.0000)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	5/5	5/5	5/5	5/5	5/5	5/5	1.000
11_Mansh	4/6	5/5	5/5	5/5	5/5	5/5	0.935
12_Marcus	5/5	5/5	5/5	5/5	5/5	5/5	1.000
13_Nat	6/6	5/5	5/5	0/5	6/6	5/5	0.844
14_StephenV2	5/5	5/5	5/5	5/5	5/5	5/5	1.000
16_Bella	5/5	5/5	5/5	5/5	5/5	5/5	1.000
17_Raquel	5/5	4/5	5/5	4/5	5/5	5/5	0.933
1_Alan	5/5	5/5	5/5	5/5	5/5	5/5	1.000
2_Alex	5/5	5/5	5/5	0/5	5/5	5/5	0.833
3_Anghad	4/5	5/5	4/5	5/5	5/5	5/5	0.933
4_Daniel	5/5	5/5	5/5	5/5	5/5	5/5	1.000
5_Georgia	5/5	4/5	5/5	5/5	5/5	5/5	0.967
6_Harry	5/5	5/5	5/5	4/5	5/5	5/5	0.967
7_Henry	5/5	5/5	4/5	5/5	5/5	5/5	0.967
8_Jestin	6/6	5/5	5/5	5/5	5/5	5/5	1.000
9_Joselyn	5/5	5/5	5/5	5/5	5/5	5/5	1.000
Stephen	3/20	18/20	15/20	17/20	18/20	18/20	0.742
Val	2/5	5/5	5/5	4/5	5/5	5/5	0.867
MEAN	0.890	0.972	0.964	0.847	0.994	0.994	0.944

Classifier: Stacking (SVM+RF+KNN+LogReg)

Params (best-effort): —

LOSO mean accuracy: 0.9497 ± 0.0689 (min 0.7500, max 1.0000)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	5/5	5/5	5/5	5/5	5/5	5/5	1.000
11_Mansh	4/6	5/5	5/5	5/5	5/5	5/5	0.935
12_Marcus	5/5	5/5	5/5	5/5	5/5	5/5	1.000
13_Nat	6/6	5/5	5/5	1/5	6/6	5/5	0.875
14_StephenV2	5/5	5/5	5/5	5/5	5/5	5/5	1.000
16_Bella	5/5	5/5	5/5	5/5	5/5	5/5	1.000
17_Raquel	5/5	4/5	5/5	5/5	5/5	5/5	0.967
1_Alan	5/5	5/5	5/5	5/5	5/5	5/5	1.000
2_Alex	5/5	5/5	5/5	1/5	5/5	5/5	0.867
3_Anghad	4/5	5/5	5/5	5/5	5/5	5/5	0.967
4_Daniel	5/5	5/5	5/5	5/5	5/5	5/5	1.000
5_Georgia	5/5	5/5	5/5	5/5	5/5	5/5	1.000
6_Harry	5/5	5/5	5/5	3/5	4/5	5/5	0.900
7_Henry	5/5	5/5	5/5	5/5	4/5	5/5	0.967
8_Jestin	6/6	5/5	5/5	5/5	5/5	5/5	1.000
9_Joselyn	5/5	5/5	5/5	5/5	5/5	5/5	1.000
Stephen	3/20	18/20	16/20	18/20	17/20	18/20	0.750
Val	2/5	5/5	5/5	4/5	5/5	5/5	0.867
MEAN	0.890	0.983	0.989	0.872	0.969	0.994	0.950